Qiqi Hu

HU-Qiqi | huqiqi.net | chelseyhu111@gmail.com

Research Interests

AI for energy systems and sustainability; battery thermal safety and thermal runaway prediction; scientific machine learning for multiphysics systems; optimization; thermal management for LIBs; generative models.

Profile

Master's student in Environmental Science and New Energy Technology at Tsinghua University (expected Dec. 2026), with an interdisciplinary background in computer science, energy engineering, and scientific machine learning. My research lies at the intersection of AI and sustainable energy systems, with current work on battery thermal safety, interpretable machine learning, and multiphysics modeling. For Ph.D. study, I aim to develop scientific machine learning and physics-informed generative methods for safety-critical energy systems, especially in battery thermal safety, multiphysics modeling, and low-carbon energy applications.

Education

Tsinghua University 2023 – 2026
M.S. in Environmental Science and New Energy Technology
Advisor: Prof. Hongda Du

Southern University of Science and Technology 2024 – 2025
Visiting Student, Department of Computer Science and Engineering

Qingdao University 2019 – 2023
B.S. in Information Security, School of Computer Science and Technology

Research Experience

Postgraduate Researcher, Tsinghua University Sep. 2023 – Present

- Guangdong Provincial Key Laboratory of Thermal Management Engineering and Materials.
- Conduct research on battery thermal management, thermal runaway safety, and AI-driven modeling for energy systems.
- Developed interpretable machine learning methods for pre-event severity prediction of lithium-ion battery thermal runaway using battery design metadata.
- Performed thermal modeling, multiphysics analysis, and data-driven optimization for battery thermal management systems under high-power operating conditions.
- Advisor: Prof. Hongda Du

Visiting Student, Southern University of Science and Technology Mar. 2024 – Dec. 2024

- Visual Intelligence and Perception Lab, Department of Computer Science and Engineering.
- Conducted research on trustworthy diffusion models and copyright/content security for AIGC systems.
- Proposed a protection framework integrating digital watermarking and latent-space perturbation for diffusion-based image-to-image generation.
- Advisor: Prof. Feng Zheng

Research Intern Dec. 2021 – Dec. 2022

- Shandong Provincial Undergraduate Innovation and Entrepreneurship Training Program (Grant No. S202111065050).
- Studied privacy-preserving computation and secure outsourced clustering for resource-constrained IoT scenarios.
- Advisor: Prof. Hanlin Zhang

Selected Publications

[J2] Safety-Oriented Pre-Event Severity Prediction of Lithium-Ion Battery Thermal Runaway Using Interpretable Machine Learning Code

First author, accepted by Process Safety and Environmental Protection (IF: 7.8)

- Proposed a metadata-driven machine learning framework for predicting thermal runaway severity without requiring calorimetry or ejection experiments.
- Developed an interpretable Severity Index and a CatBoost-based classifier, achieving 81% accuracy and 100% recall for high-severity cells.
- Used SHAP-based interpretation and sensitivity analysis to identify key design factors and evaluate engineering robustness.

[J1] Design and Multi-Objective Optimization of an Efficient UAV Battery Thermal Management System Using a PCM–Air Synergistic Cooling Strategy

First author, accepted by Applied Thermal Engineering (IF: 6.9)

- Designed a hybrid battery thermal management system combining phase change materials and air cooling for high-power UAV operating conditions.
- Conducted multiphysics simulations and parametric analysis, followed by entropy-weighted TOPSIS for multi-objective optimization.
- Reduced maximum battery temperature by 43.3% to below 46.8 °C, with only a 22.2 wt% increase in system mass.

Selected Research Projects

[P2] Copyright Protection for Diffusion-Based Image-to-Image Generation Code

- Proposed a dual-protection framework that combines digital watermarking and latent-space adversarial perturbations for diffusion models.
- Achieved robust and transferable copyright protection without model fine-tuning while preserving image quality and reliable watermark extraction.

[P1] Privacy-Preserving Shared Nearest Neighbor Clustering for the Internet of Things Preprint

- Developed a cloud-assisted privacy-preserving SNN clustering framework for resource-constrained IoT devices using orthogonal matrix transformation.
- Reduced local computation time from 10.859 s to 1.183 s while preserving clustering performance on real-world datasets.

Honors and Awards

- **Typical Graduate (Ranked 1st)**, School of Computer Science and Technology, Qingdao University, 2023
- **Qingdao University Star Award**, highest university-wide student honor, 2023
- Honorable Mention, International Mathematical Contest in Modeling (MCM), 2021

Technical Skills

Programming	Python, MATLAB, C, LaTeX
ML Libraries	PyTorch, TensorFlow, scikit-learn
Simulation / Tools	COMSOL, Linux, VS Code, Anaconda
Languages	Chinese (native), English (fluent)

Last updated: April 7, 2026